

Decent Coaching Classes

VIII Praneet

Maths Revision Worksheet – Class VIII (NCERT)

Chapters: Squares, Square Roots & Cubes • Exponents • Quadrilaterals • Playing with Numbers • We Distribute, Yet Things Multiply

Show all working. Answer key is given at the end.

SECTION A – SQUARES, SQUARE ROOTS AND CUBES (Ch 1)

1. Write the units digit of the square of each number.
(a) 67 (b) 354 (c) 1238 (d) 9999 (e) 25640
2. Give a reason why none of the following is a perfect square.
(a) 2563 (b) 7778 (c) 48000 (d) 150
3. Find the squares of the following numbers.
(a) 16 (b) 35 (c) 55 (d) 110 (e) 2000
4. Which of the following are perfect squares?
(a) 169 (b) 484 (c) 1089 (d) 2025 (e) 1782
5. How many natural numbers lie between the squares of the following numbers?
(a) 12 and 13 (b) 20 and 21 (c) 50 and 51
6. Using the property of squares, find the following products.
(a) 29×31 (b) 59×61 (c) 99×101
7. Without adding, find the sum of the following.
(a) $1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17$
(b) $1 + 3 + 5 + 7 + \dots + 25$ (13 terms)
8. Which digits can never be the units digit of a perfect square?
9. Find the square root using prime factorisation.
(a) 441 (b) 1764 (c) 3600
10. Find the square root using the long division method.
(a) 2209 (b) 5329 (c) 24336

11. Without finding the square root, state the number of digits in the square root of each number.
 (a) 5476 (b) 123201 (c) 9801 (d) 1000000
12. Check whether the following numbers are perfect cubes.
 (a) 1728 (b) 2744 (c) 3000 (d) 9261 (e) 4500
13. (a) Find the smallest natural number by which 108 must be multiplied to get a perfect cube. (b) Find the smallest natural number by which 675 must be divided to get a perfect cube.
14. (a) Is 180 a perfect square? If not, find the smallest natural number by which it must be multiplied to make it a perfect square. (b) By which smallest number should 1250 be divided to get a perfect square?
15. (a) The area of a square garden is 1296 m^2 . Find the length of its side and the length of fencing needed to go around it. (b) A cube-shaped water tank has a volume of 3375 m^3 . Find the length of its edge.

SECTION B – POWERS AND EXPONENTS (Ch 2)

1. Tick (✓) the correct option.
 (a) The value of $(-3)^2 \times (-2)^3$ is
 (i) -72 (ii) 72 (iii) -216 (iv) 36
 (b) The exponential form of $a \times a \times b \times b \times b \times c$ is
 (i) a^2b^3c (ii) a^3b^2c (iii) $a^2b^2c^3$ (iv) a^3b^3c
 (c) The expression with a negative value is
 (i) $(-2)^4$ (ii) $(-3)^2$ (iii) $(-5)^3$ (iv) $(-1)^6$
2. Express in exponential form.
 (a) $5 \times 5 \times 5 \times 5 \times 7 \times 7$ (b) $x \times x \times y \times y \times y \times y$ (c) $6 \times 6 \times 6 \times a \times a \times b$
3. Express each number as a product of powers of its prime factors.
 (a) 360 (b) 1176 (c) 2250
4. Evaluate.
 (a) $(-2)^5 \times 3^2$ (b) $(-1)^{75} \times (-1)^{40}$
 (c) $(-1)^{12} + (-1)^{31} + (-1)^{100}$ (d) $4^2 \times (-3)^3$
5. Identify the greater number in each of the following.
 (a) 2^9 or 9^2 (b) 5^3 or 3^5

6. Write each as a power of a power in at least two different ways.
 (a) 3^8 (b) 7^{12} (c) 2^{20}
7. Write the equivalent form with a positive exponent and simplify.
 (a) 3^{-4} (b) $(-5)^{-3}$ (c) 10^{-6} (d) 2^{-5}
8. Simplify and write the answer in exponential form.
 (a) $7^0 \times a^0 \times b^0$ (b) $(x^2y^3)^4$
 (c) $9^6 \div (3^5 \times 3^3)$ (d) $12^{10} \div (3^8 \times 4^8)$
9. A strip of paper is 96 cm long. It is folded in half 5 times. What is the length of the folded strip?
10. (a) A restaurant offers 6 starters and 5 main courses. How many different meals of one starter and one main course are possible? (b) A lock needs a 3-digit code, and each digit can be any number from 0 to 9. How many different codes are possible?
11. Express the following numbers in standard form.
 (a) 0.000000048 (b) 0.0000000000312
 (c) 7,500,000,000 (d) 0.00000604
12. Express the following in the usual form.
 (a) 6.2×10^{-6} (b) 3.45×10^{-9}
 (c) 9.01×10^{-4} (d) 2.5×10^7
13. The diameter of a virus is 0.00000012 m. Express it in standard form.

SECTION C – QUADRILATERALS (Ch 4)

1. State True or False.
 (a) The diagonals of a rhombus are equal. (b) The diagonals of a rectangle bisect each other.
 (c) Every square is a rhombus. (d) Every parallelogram is a rectangle.
 (e) A kite has a pair of parallel sides. (f) Opposite angles of a parallelogram are equal.
2. In rectangle ABCD, the diagonals AC and BD intersect at O. Given $\angle OAB = 38^\circ$, find:
 (i) $\angle OBA$ (ii) $\angle AOB$ (iii) $\angle BOC$ (iv) $\angle ACB$
3. The diagonals of rectangle EFGH are $EG = (5x + 3)$ cm and $FH = (3x + 13)$ cm. Find the value of x, the length of each diagonal and the length of EO (O is the point of intersection).

4. A quadrilateral has two diagonals, each 20 cm long, which bisect each other at O. What can you conclude about the quadrilateral and its sides?
5. The angles of a quadrilateral are $(3x)^\circ$, $(2x + 10)^\circ$, $(4x - 20)^\circ$ and 100° . Find x and all four angles.
6. Two angles of a quadrilateral are 65° and 95° , and the remaining two are $(3x)^\circ$ and $(2x)^\circ$. Find the remaining angles.
7. In a quadrilateral, the largest angle is three times the smallest angle and the other two angles are 70° each. Find all four angles.
8. In quadrilateral PQRS, $\angle P = \angle Q$ and $\angle R = \angle S$. Also $\angle P$ is double of $\angle R$. Find all four angles.
9. In parallelogram ABCD, $AB = (5a - 3)$ cm, $CD = (2a + 9)$ cm, $AD = (4b + 1)$ cm and $BC = (3b + 6)$ cm. Find a , b and the lengths of AB and AD .
10. In parallelogram ABCD, $\angle A = 110^\circ$. Find $\angle B$, $\angle C$ and $\angle D$.
11. PQRS is a rhombus with $\angle PQR = 74^\circ$. Find $\angle QRS$ and $\angle QSR$.
12. LMNO is a rhombus with $\angle LMN = 84^\circ$. Find $\angle MNO$ and $\angle NOL$.
13. If one base angle of an isosceles trapezium is 65° , find the other three angles.
14. In trapezium ABCD, $AB \parallel DC$, $\angle A = 70^\circ$ and $\angle B = 55^\circ$. Find $\angle C$ and $\angle D$.
15. In quadrilateral ABCD, $\angle A = 125^\circ$, $\angle B = 95^\circ$, $\angle C = 85^\circ$ and $\angle D = 55^\circ$. Can you say that ABCD is a trapezium? Give a reason.
16. Is a rectangle a kite? Give reasons.
17. Two equilateral triangles, each of side 6 cm, are joined along a common side. What quadrilateral is formed? Find all its sides and angles.

SECTION D – PLAYING WITH NUMBERS (Ch 5)

1. The sum of four consecutive numbers is 130. Find the numbers.
2. Suppose r is the greatest of five consecutive numbers. Describe the other four numbers in terms of r .
3. The middle number in a sequence of 5 consecutive odd numbers is $(4t + 1)$. Express the other four numbers in terms of t .
4. The sum of three consecutive multiples of 9 is 351. Find the multiples.
5. Write three algebraic expressions that always give (a) an odd number, (b) an even number.
6. Raj claims, "If you add any two numbers that leave a remainder of 4 when divided by 8, the sum will always be a multiple of 8." Is Raj's claim true? Justify using algebra.
7. A number A leaves remainder 5 when divided by 7, and a number B (smaller than A) leaves remainder 4. Without performing division, find the remainder when each of the following is divided by 7.
(a) $A + B$ (b) $A - B$ (c) $A \times B$
8. Using divisibility tests, answer the following.
(a) Which are divisible by 3? 456, 7291, 2034, 58171
(b) Which are divisible by 9? 3456, 7821, 5432, 12345
(c) Which are divisible by 4? 7316, 5042, 98124, 3570
(d) Which are divisible by 11? 82918, 30517, 4125, 90729
(e) Which are divisible by 5? Which by 10? 4205, 9830, 7318, 6000
9. If the division $N \div 5$ leaves a remainder of 3, what might be the ones digit of N ?
10. (a) If $52z8$ is divisible by 3, where z is a digit, find the possible values of z . (b) If $7y4$ is divisible by 9, where y is a digit, find the value of y .
11. Find the digital root of the following numbers.
(a) 9876 (b) 45672
12. Between the numbers 600 and 700, which numbers have the digital root 4?

13. Take 29 as a number. Generate a sequence of 10 numbers by repeatedly adding 12. What are the digital roots of this sequence? State the pattern you observe.

14. Fill the table and then answer.

Number	Parity (E/O)	Sum of digits	Digital root	Parity of digital root
124				
345				
568				
791				

(a) Does the parity of a number determine the parity of its digital root?

(b) Give one example of an even number with an odd digital root and one odd number with an even digital root.

15. The digital root of a number is 6. What will be the digital root of the number that is 14 more than it?

SECTION E – WE DISTRIBUTE, YET THINGS MULTIPLY (Ch 6)

1. Expand using the distributive property.

(a) $(\frac{3}{4})(8a - 4b + 12)$

(b) $(\frac{5}{6})(12x + 18y - \frac{6}{5})$

2. Multiply.

(a) $3xy$ by $(2x^2 - 5xy)$

(b) $-4a^2b$ by $(3a - 2b + 5)$

3. Find the products.

(a) $(3a + 5)(2a + 7)$

(b) $(4x^2 + 3)(5x^2 - 2)$

4. Find the products.

(a) $(x - 3)(x^2 + 3x + 9)$

(b) $(a + 2)(a^2 - 2a + 4)$

5. Simplify: $(2x + y)(x - 3y) - 2x(x - 5y)$

6. Find the value of $2x^3 \times (-\frac{1}{3}xy^2) \times 6yz^2$, when $x = 1$, $y = 2$ and $z = 3$.

7. Write the product in one line.

(a) 63×11

(b) 458×11

(c) 7215×11

(d) 47×101

(e) 625×101

8. Use the distributive property to find the following products.

(a) 456×99

(b) 7321×999

(c) 2358×1001

9. Expand using an identity.
 (a) $(5a + 4)^2$ (b) $(m - 9)^2$ (c) $(-2x + 7)^2$
10. Using identities, evaluate.
 (a) 73^2 (b) 197^2 (c) 53×47 (d) 105×95
11. Using the identity $a^2 - b^2 = (a + b)(a - b)$, evaluate.
 (a) $85^2 - 15^2$ (b) $632^2 - 368^2$
12. Find the products.
 (a) $(3a^2 - 2b)(3a^2 + 2b)$ (b) $(2p + 3q)(2p - 3q)$
13. Find the continued products.
 (a) $(x + 3)(x - 3)(x^2 + 9)$ (b) $(2a - 1)(2a + 1)(4a^2 + 1)$
14. Find the value of x.
 (a) $8x = 45^2 - 37^2$ (b) $12x = 58^2 - 46^2$
15. Using identities, evaluate.
 (a) $(76 \times 76 - 24 \times 24) / (76 - 24)$ (b) $(53 \times 53 - 17 \times 17) / (53 + 17)$
16. Tanvi simplified $3(y - 2) + 4(y + 5) = 3y - 2 + 4y + 5 = 7y + 3$. Is there a mistake? If yes, explain and write the correct expression.
17. Verify that $(2x + 3y)(2x - 3y)(4x^2 + 9y^2) = 16x^4 - 81y^4$ for $x = 1$ and $y = 1$.

— End of Worksheet —

ANSWER KEY

Maths Revision Worksheet – VIII Praneeet

Section A – Squares, Square Roots and Cubes

1. (a) 9 (b) 6 (c) 4 (d) 1 (e) 0
2. (a) ends in 3 (b) ends in 8 (c) three (odd number of) zeros at the end (d) one (odd number of) zero at the end. A perfect square never ends in 2, 3, 7, 8 or an odd number of zeros.
3. (a) 256 (b) 1225 (c) 3025 (d) 12100 (e) 4000000
4. $169 = 13^2$, $484 = 22^2$, $1089 = 33^2$, $2025 = 45^2$ are perfect squares. 1782 is not (ends in 2).
5. Between n^2 and $(n + 1)^2$ there are $2n$ numbers: (a) 24 (b) 40 (c) 100
6. Using $(a - 1)(a + 1) = a^2 - 1$: (a) $30^2 - 1 = 899$ (b) $60^2 - 1 = 3599$ (c) $100^2 - 1 = 9999$
7. Sum of first n odd numbers $= n^2$: (a) 9 terms, $9^2 = 81$ (b) 13 terms, $13^2 = 169$
8. 2, 3, 7 and 8
9. (a) $441 = 3^2 \times 7^2$, so $\sqrt{441} = 21$ (b) $1764 = 2^2 \times 3^2 \times 7^2$, so $\sqrt{1764} = 42$ (c) $3600 = 2^4 \times 3^2 \times 5^2$, so $\sqrt{3600} = 60$
10. (a) 47 (b) 73 (c) 156
11. (a) 2 digits (b) 3 digits (c) 2 digits (d) 4 digits (1000000 has 7 digits, and $(7 + 1)/2 = 4$)
12. (a) Yes (12^3) (b) Yes (14^3) (c) No (d) Yes (21^3) (e) No
13. (a) $108 = 2^2 \times 3^3$; multiply by 2 to get $216 = 6^3$. (b) $675 = 3^3 \times 5^2$; divide by 25 to get $27 = 3^3$.
14. (a) $180 = 2^2 \times 3^2 \times 5$. Not a perfect square; multiply by 5 to get $900 = 30^2$. (b) $1250 = 2 \times 5^4$; divide by 2 to get $625 = 25^2$.
15. (a) Side $= \sqrt{1296} = 36$ m; fencing $= 4 \times 36 = 144$ m. (b) Edge $= \sqrt[3]{3375} = 15$ m.

Section B – Powers and Exponents

1. (a) (i) -72 (b) (i) a^2b^3c (c) (iii) $(-5)^3$
2. (a) $5^4 \times 7^2$ (b) x^2y^4 (c) 6^3a^2b
3. (a) $2^3 \times 3^2 \times 5$ (b) $2^3 \times 3 \times 7^2$ (c) $2 \times 3^2 \times 5^3$
4. (a) $-32 \times 9 = -288$ (b) $(-1)^{115} = -1$ (c) $1 - 1 + 1 = 1$ (d) $16 \times (-27) = -432$
5. (a) $2^9 = 512 > 9^2 = 81$, so 2^9 is greater. (b) $3^5 = 243 > 5^3 = 125$, so 3^5 is greater.
6. (a) $(3^2)^4$, $(3^4)^2$ (b) $(7^3)^4$, $(7^6)^2$, $(7^2)^6$ (c) $(2^4)^5$, $(2^5)^4$, $(2^2)^{10}$, $(2^{10})^2$
7. (a) $1/3^4 = 1/81$ (b) $1/(-5)^3 = -1/125$ (c) $1/10^6 = 1/1000000$ (d) $1/2^5 = 1/32$
8. (a) 1 (b) x^8y^{12} (c) $3^{12} \div 3^8 = 3^4 (= 81)$ (d) $12^{10} \div 12^8 = 12^2 (= 144)$
9. $96 \div 2^5 = 96 \div 32 = 3$ cm
10. (a) $6 \times 5 = 30$ meals (b) $10 \times 10 \times 10 = 10^3 = 1000$ codes
11. (a) 4.8×10^{-8} (b) 3.12×10^{-11} (c) 7.5×10^9 (d) 6.04×10^{-6}
12. (a) 0.0000062 (b) 0.00000000345 (c) 0.000901 (d) 25,000,000
13. 1.2×10^{-7} m

Section C – Quadrilaterals

1. (a) False (b) True (c) True (d) False (e) False (f) True
2. $OA = OB$, so $\angle OBA = 38^\circ$. (i) $\angle OBA = 38^\circ$ (ii) $\angle AOB = 180^\circ - 76^\circ = 104^\circ$ (iii) $\angle BOC = 180^\circ - 104^\circ = 76^\circ$ (iv) $\angle ACB = 90^\circ - 38^\circ = 52^\circ$
3. Diagonals of a rectangle are equal: $5x + 3 = 3x + 13$, so $x = 5$. $EG = FH = 28$ cm, $EO = 14$ cm.
4. It is a rectangle (a parallelogram whose diagonals are equal). Opposite sides are equal and parallel, and every angle is 90° .
5. $3x + 2x + 10 + 4x - 20 + 100 = 360 \Rightarrow 9x + 90 = 360 \Rightarrow x = 30$. Angles: 90° , 70° , 100° , 100° .
6. $65 + 95 + 3x + 2x = 360 \Rightarrow 5x = 200 \Rightarrow x = 40$. Remaining angles: 120° and 80° .
7. Let smallest $= s$: $s + 3s + 70 + 70 = 360 \Rightarrow s = 55^\circ$. Angles: 55° , 70° , 70° , 165° .

8. $\angle P = \angle Q = 2\angle R$, $\angle R = \angle S$. So $2R + 2R + R + R = 360^\circ \Rightarrow R = 60^\circ$. $\angle P = \angle Q = 120^\circ$, $\angle R = \angle S = 60^\circ$.
9. $5a - 3 = 2a + 9 \Rightarrow a = 4$; $4b + 1 = 3b + 6 \Rightarrow b = 5$. $AB = 17$ cm, $AD = 21$ cm.
10. $\angle B = 70^\circ$, $\angle C = 110^\circ$, $\angle D = 70^\circ$ (opposite angles equal, adjacent angles add up to 180°).
11. $\angle QRS = 180^\circ - 74^\circ = 106^\circ$. $\angle PSR = \angle PQR = 74^\circ$ and diagonal SQ bisects it, so $\angle QSR = 37^\circ$.
12. $\angle MNO = 180^\circ - 84^\circ = 96^\circ$; $\angle NOL = \angle LMN = 84^\circ$.
13. Base angles are equal: 65° and 65° ; the other two angles are each $180^\circ - 65^\circ = 115^\circ$.
14. $\angle D = 180^\circ - 70^\circ = 110^\circ$ and $\angle C = 180^\circ - 55^\circ = 125^\circ$ (co-interior angles between parallel sides).
15. Yes. $\angle B + \angle C = 95^\circ + 85^\circ = 180^\circ$, so $AB \parallel DC$ (co-interior angles are supplementary). A quadrilateral with one pair of parallel sides is a trapezium.
16. No. A kite has two pairs of equal adjacent sides. In a rectangle, opposite sides are equal but adjacent sides are not equal (unless it is a square).
17. A rhombus. All sides are 6 cm. Angles: 60° , 120° , 60° , 120° .

Section D – Playing with Numbers

1. $x + (x + 1) + (x + 2) + (x + 3) = 130 \Rightarrow 4x + 6 = 130 \Rightarrow x = 31$. Numbers: 31, 32, 33, 34.
2. $r - 4$, $r - 3$, $r - 2$, $r - 1$ (and r)
3. $4t - 3$, $4t - 1$, $4t + 1$, $4t + 3$, $4t + 5$
4. $n + (n + 9) + (n + 18) = 351 \Rightarrow 3n + 27 = 351 \Rightarrow n = 108$. Multiples: 108, 117, 126.
5. (Any valid answers.) (a) Odd: $2n + 1$, $2n - 1$, $4n + 3$ (b) Even: $2n$, $4n$, $6n + 2$
6. True. Numbers are $8a + 4$ and $8b + 4$. Sum = $8a + 8b + 8 = 8(a + b + 1)$, which is a multiple of 8.
7. (a) $5 + 4 = 9$, remainder 2 (b) $5 - 4 = 1$, remainder 1 (c) $5 \times 4 = 20$, remainder 6
8. (a) 456 and 2034 (b) 3456 and 7821 (c) 7316 and 98124 (d) 82918 and 4125 (e) Divisible by 5: 4205, 9830, 6000; divisible by 10: 9830, 6000
9. 3 or 8
10. (a) $5 + 2 + z + 8 = 15 + z$ must be a multiple of 3, so $z = 0, 3, 6$ or 9. (b) $7 + y + 4 = 11 + y$ must be a multiple of 9, so $y = 7$.
11. (a) $9 + 8 + 7 + 6 = 30 \rightarrow 3$ (b) $4 + 5 + 6 + 7 + 2 = 24 \rightarrow 6$
12. 607, 616, 625, 634, 643, 652, 661, 670, 679, 688, 697
13. Sequence: 29, 41, 53, 65, 77, 89, 101, 113, 125, 137. Digital roots: 2, 5, 8, 2, 5, 8, 2, 5, 8, 2. The digital roots repeat in the pattern 2, 5, 8 (adding 12 adds 3 to the digital root each time).
14. Table (Number: Parity, Sum, Digital root, Parity of DR): 124: E, 7, 7, O; 345: O, 12, 3, O; 568: E, 19, 1, O; 791: O, 17, 8, E. (a) No. (b) Even number with odd digital root: 124 (digital root 7). Odd number with even digital root: 791 (digital root 8).
15. Digital root of 14 is 5, so $6 + 5 = 11 \rightarrow$ digital root 2.

Section E – We Distribute, Yet Things Multiply

1. (a) $6a - 3b + 9$ (b) $10x + 15y - 1$
2. (a) $6x^3y - 15x^2y^2$ (b) $-12a^3b + 8a^2b^2 - 20a^2b$
3. (a) $6a^2 + 31a + 35$ (b) $20x^4 + 7x^2 - 6$
4. (a) $x^3 - 27$ (b) $a^3 + 8$
5. $2x^2 - 6xy + xy - 3y^2 - 2x^2 + 10xy = 5xy - 3y^2$
6. The expression simplifies to $-4x^4y^3z^2$. For $x = 1$, $y = 2$, $z = 3$: $-4 \times 1 \times 8 \times 9 = -288$
7. (a) 693 (b) 5038 (c) 79365 (d) 4747 (e) 63125
8. (a) $456 \times (100 - 1) = 45600 - 456 = 45144$ (b) $7321 \times (1000 - 1) = 7321000 - 7321 = 7313679$ (c) $2358 \times (1000 + 1) = 2358000 + 2358 = 2360358$
9. (a) $25a^2 + 40a + 16$ (b) $m^2 - 18m + 81$ (c) $4x^2 - 28x + 49$
10. (a) $(70 + 3)^2 = 5329$ (b) $(200 - 3)^2 = 38809$ (c) $(50 + 3)(50 - 3) = 2491$ (d) $(100 + 5)(100 - 5) = 9975$
11. (a) $(85 - 15)(85 + 15) = 70 \times 100 = 7000$ (b) $(632 - 368)(632 + 368) = 264 \times 1000 = 264000$

12. (a) $9a^4 - 4b^2$ (b) $4p^2 - 9q^2$
13. (a) $(x^2 - 9)(x^2 + 9) = x^4 - 81$ (b) $(4a^2 - 1)(4a^2 + 1) = 16a^4 - 1$
14. (a) $8x = (45 - 37)(45 + 37) = 8 \times 82$, so $x = 82$ (b) $12x = (58 - 46)(58 + 46) = 12 \times 104$, so $x = 104$
15. (a) $(76 + 24)(76 - 24) / (76 - 24) = 76 + 24 = 100$ (b) $(53 + 17)(53 - 17) / (53 + 17) = 53 - 17 = 36$
16. Yes. 3 and 4 were not multiplied with -2 and 5 . Correct: $3(y - 2) + 4(y + 5) = 3y - 6 + 4y + 20 = 7y + 14$.
17. $\text{LHS} = (2 + 3)(2 - 3)(4 + 9) = 5 \times (-1) \times 13 = -65$. $\text{RHS} = 16 - 81 = -65$. $\text{LHS} = \text{RHS}$, verified.